

Calculus II Lesson 18

1) $\sum_{n=1}^{\infty} \frac{(-1)^n}{2^n}$ converges absolutely, converges conditionally, or diverges?

Note: $\left| \frac{(-1)^n}{2^n} \right| = \frac{1}{2^n} = \left(\frac{1}{2} \right)^n$

a) Use the Alternating Series Test to explain why the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{2^n}$ converges or diverges.

b) Use the Geometric Series Test to explain why the series $\sum_{n=1}^{\infty} \left| \frac{(-1)^n}{2^n} \right| = \sum_{n=1}^{\infty} \left(\frac{1}{2} \right)^n$ converges or diverges.

c) Does the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{2^n}$ converge absolutely or conditionally?

2) $\sum_{n=1}^{\infty} \frac{(-1)^n}{n!}$ converges absolutely, converges conditionally, or diverges?

Note: $\left| \frac{(-1)^n}{n!} \right| = \frac{1}{n!}$

a) Use the Alternating Series Test to explain why the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n!}$ converges or diverges.

b) Use the Basic Comparison Test to explain why the series $\sum_{n=1}^{\infty} \left| \frac{(-1)^n}{n!} \right| = \sum_{n=1}^{\infty} \frac{1}{n!}$ converges or diverges.

Hint: $\frac{1}{n!} < (0.9)^n$ for $n = 2, 3, 4, \dots$

c) Does the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n!}$ converge absolutely or conditionally?

3) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}}$ converges absolutely, converges conditionally, or diverges?

Note: $\left| \frac{(-1)^{n+1}}{\sqrt{n}} \right| = \frac{1}{\sqrt{n}} = \frac{1}{n^{1/2}}$

a) Use the Alternating Series Test to explain why the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}}$ converges or diverges.

b) Use the p -series Test to explain why the series $\sum_{n=1}^{\infty} \left| \frac{(-1)^{n+1}}{\sqrt{n}} \right| = \sum_{n=1}^{\infty} \frac{1}{n^{1/2}}$ converges or diverges.

c) Does the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$ converge absolutely or conditionally?

4) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} n^2}{(n+1)^2}$ converges or diverges?

Note: $\left| \frac{(-1)^{n+1} n^2}{(n+1)^2} \right| = \frac{n^2}{(n+1)^2}$

a) Find $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{n^2}{(n+1)^2} = \underline{\hspace{1cm}} ?$

Hint: Use L'Hopital's Rule

b) Use the n th Term Test to explain why the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} n^2}{(n+1)^2}$ diverges.

5) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} n}{n^3 - 5}$ converges or diverges?

Note: $\left| \frac{(-1)^{n+1} n}{n^3 - 5} \right| = \frac{n}{n^3 - 5}$

a) Use the Alternating Series Test to explain why the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} n}{n^3 - 5}$ converges or diverges.

b) Explain why the series $\sum_{n=1}^{\infty} \left| \frac{(-1)^{n+1} n}{n^3 - 5} \right| = \sum_{n=1}^{\infty} \frac{n}{n^3 - 5}$ converges or diverges.

Hint: Use the Limit Comparison Test with $a_n = \frac{n}{n^3 - 5}$ and $b_n = \frac{n}{n^3} = \frac{1}{n^2}$

c) Does the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} n}{n^3 - 5}$ converge absolutely or conditionally?

6) $\sum_{n=1}^{\infty} \frac{(-1)^n}{(2n+1)!}$ converges absolutely, converges conditionally, or diverges?

Note: $\left| \frac{(-1)^n}{(2n+1)!} \right| = \frac{1}{(2n+1)!}$

a) Use the Alternating Series Test to explain why the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{(2n+1)!}$ converges or diverges.

b) Use the Basic Comparison Test to explain why the series $\sum_{n=1}^{\infty} \left| \frac{(-1)^n}{(2n+1)!} \right| = \sum_{n=1}^{\infty} \frac{1}{(2n+1)!}$ converges or diverges.

Hint: $\frac{1}{(2n+1)!} < \frac{1}{n!}$

$$\sum_{n=1}^{\infty} \frac{1}{(2n+1)!} < \sum_{n=1}^{\infty} \frac{1}{n!}$$

In a video in Lesson 15, $\sum_{n=1}^{\infty} \frac{1}{n!}$ was shown to converge by B.C.T.

c) Does the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{(2n+1)!}$ converge absolutely or conditionally?

